

Alvin Tan

Research Assistant. Dept. Electr. Eng. & Comp. Sci., UC Berkeley.
alverino@berkeley.edu. 515-708-4666. <https://www.linkedin.com/in/alvin-tan-10533012a>

Education

UC Berkeley -- PhD Candidate. EECS with Prof. Prabal Dutta Expected Dec 2026 | GPA 3.95/4.00
Northwestern University -- BS in Comp Eng, Econ, and Math May 2020 | GPA 3.99/4.00

Honors and Awards

2026	LBNL Director's Award in Exceptional Scientific Achievement
2026	UC Berkeley's EECS Evergreen Award for Undergraduate Researcher Mentoring
2023	UC Berkeley's EECS Outstanding TA Award (Top 1%)
2023	UC Berkeley's Outstanding Graduate Student Instructor Award (Top 10%)
2022	Johns Hopkins University Applied Physics Lab's Bravo Award
2022	NSF GRFP Honorable Mention
2020	Northwestern's ECE Outstanding Graduating Senior
2018-2020	Tau Beta Pi Honor Society
2017-2020	Eta Kappa Nu Honor Society
2016-2020	Northwestern's Dean's List with High Honors
2018	Goldwater Scholarship Honorable Mention
2017	Northwestern's Eshbach Award for Scholarship and Leadership

Publications and Exhibitions

Alvin Tan, U. Gopi, P. Dutta. *Data Trickle: A Study on Reducing Transmissions for Live, Low-Data-Rate Data Streams*. Submitted to SenSys 2027 (ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems).

Alvin Tan, D. Gerber, P. Dutta. *Less is the New More: On Reducing Panel Capacity Requirements using PV- and Battery-Supported DC Microgrids*. *Energies* 2026, 19, 4051/1-45. DOI: 10.3390/en19174051.

Alvin Tan, D. Gerber, P. Dutta. *Not too big, not too small... Towards Minimum Battery Sizing for Capacity-Limited DC Microgrids*. 2026 IEEE Eighth International Conference on DC Microgrids (ICDCM), pp. 1-6. DOI: 10.1109/ICDCM67837.2026.11594463.

D. Jackson, B. Nordman, L. Paul, **Alvin Tan**. *Price-based Demand Flexibility with OpenADR 3: Inexpensive, Simple, and Ready*. Poster and hardware demo presented in the 2025 California Demand Flexibility Summit, Hackathon at Port Labs in Oakland, and at SPAN.IO Inc.

Alvin Tan, P. Dutta. *Good, but Not That Good: An Honestly-Noisy Visualization of Low-Fidelity Data Streams*. HumanSys '25: Proceedings of the 3rd International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems. pp 98 – 103. DOI: 10.1145/3722570.3726882.

Alvin Tan. *A Sum of Parts: Contemplating the Value of Life and Gentleness through Technology*. Short film presented in the Arts in Robotics gallery exhibition at 2025 IEEE International Conference on Robotics & Automation (ICRA).

S. Pohland*, **Alvin Tan***, P. Dutta, C. Tomlin. *Stranger Danger! Identifying and Avoiding Unpredictable Pedestrians in RL-based Social Robot Navigation*. 2024 IEEE International Conference on Robotics and Automation (ICRA). pp. 15217-15224, DOI: 10.1109/ICRA57147.2024.10610413. (*co-first authorship)

J.L. Watson, T. Despres, **Alvin Tan**, S. Patil, P. Dutta, R. Popa. *Nebula: A Privacy-First Platform for Data Backhaul*. 2024 IEEE Symposium on Security and Privacy (SP). pp. 3184-3202. DOI: 10.1109/SP54263.2024.00092.

T. Despres, S. Patil, **Alvin Tan**, J.L. Watson, P. Dutta. *Where the Sidewalk Ends: Privacy of Opportunistic Backhaul*. EuroSec '22: Proceedings of the 15th European Workshop on Systems Security. pp. 1 – 7. DOI: 10.1145/3517208.3523757.

N. Drenkow, **Alvin Tan**, C. Ashcraft, K. Karra. *Context-Adaptive Deep Neural Networks via Bridge-Mode Connectivity*. NeurIPS 2022 Machine Learning Safety Workshop. pp. 1-8. DOI: 10.48550/arXiv.2211.15436.

Teaching and Mentoring

UC Berkeley Electrical Engineering and Computer Science Department 2024-present
Graduate Student Researcher

- Provided research guidance and mentorship for four undergraduate students

UC Berkeley Electrical Engineering and Computer Science Department 2023-2024
Graduate Peer Mentor

- Provided recommendations on course selection, advisor assignments, and research projects for prospective graduate students during Visit Days and the first year of PhD

UC Berkeley Electrical Engineering and Computer Science Department 2022
Embedded Systems (EECS 149) Lab Instructor

- Presented course material and facilitated three embedded-systems lab sections per week for a class of 80 undergraduate and graduate students

Eta Kappa Nu Honor Society 2019-2020
Freshman Peer Mentor

- Provided recommendations on course selection, advisor assignments, and research projects for freshmen in the Electrical and Computer Engineering department

Northwestern Academic Support and Learning Advancement 2017-2020
Microeconomics Peer Facilitator

- Presented course material and created practice problems to review concepts for up to ten university students two hours every week

NU Tutors 2016-2020
Calculus AB and Physics B Tutor

- Reviewed calculus and physics concepts with local high school students to prepare for homeworks, tests, and AP exams

Northwestern Electrical Engineering and Computer Science Department 2018-2019
Computer Systems (CS 213) Peer Mentor

- Reviewed computer systems topics with university students and provided advice and troubleshooting on course projects

Research Projects

Power-Efficient Data Upload Strategies for Low-Fidelity Data Streams 2023-present
Prof. Prabal Dutta (UC Berkeley)

- Reviewed up-to-date literature articles on data compression, handling data loss in streaming, perception-less compression, and data visualization for live data streaming of user applications
- Identified a subset of sensing systems characterized by live-but-sparse data updates
- Evaluated a broad set of data-reduction and upload strategies for such systems
- Mentored 3 undergraduates on conducting system-level experiments, with 2 papers in progress
- Achieved up to 2x higher compression efficiency compared to state-of-practice, resulting in estimated 16-25% longer sensor lifetimes for select industrial and residential sensor systems
- Analyzed 21 compression algorithms across 12 publications, categorized them into 12 groups using common operations, and used that to automatically optimize compression of 15 timeseries datasets

Evaluating General Public's Interpretation of Low-Fidelity Data Streams 2023-2025
Prof. Prabal Dutta (UC Berkeley)

- Articulated and quantified user-perceived error and stability of low-fidelity data streams from a hypothetical temperature sensor deployment on university campus
- Demonstrated user misinterpretation of error-triggered data updates and the ability for a conceptual "noisy sensor" to correct the heuristics used when interpreting data

Towards Mega-Pixel Imagers in Fiber-Form-Factor Computers July-Dec 2023
Prof. Prabal Dutta (UC Berkeley)

- Explored system architectures for concurrently collecting, compressing, and storing megapixel-sized images using heavily resource-constrained microcontrollers

Privacy-Preserving, World-Scale, Opportunistic Networking System 2021-2024

Prof. Prabal Dutta (UC Berkeley)

- Designed and evaluated a spam-resistant and scalable system to securely upload sensor data to the cloud through nearby smartphones while preserving user privacy

Uncertainty-Aware Social Robot Navigation in Pedestrian Spaces 2021-2024

Prof. Prabal Dutta (UC Berkeley)

- Reviewed literature on current research on social robot navigation, proxemics, pathfinding, and uncertainty quantification, and identified gaps in existing work on reducing robot failures in situations with unforeseen pedestrian movement
- Designed and implemented a training procedure to embed uncertainty-awareness into reinforcement-learning-based social robot navigation policies
- Trained a reinforcement-learning model (100k trainable parameters) using imitation learning and custom curriculum training, reduced human-robot collisions by 80% in hard scenarios
- Integrated the model into a 2-wheeled robot with computer vision (YOLO on an RGB-D camera) and motor controls, and demonstrated that learned behavior transfers from simulation to the real-world
- Tackled practical pedestrian tracking challenges: computed coordinate transformations between the camera and robot, mitigated measurement latency with dead-reckoning, tracked pedestrians in 2D to reconcile redundant measurements, and controlled motors at a frequency higher than the camera

Electrical Load Classification for Residential Energy Monitoring 2020-2021

Prof. Prabal Dutta (UC Berkeley)

- Evaluated existing residential energy monitoring efforts and developed machine learning solutions to enable large-scale deployment of power monitoring submeters

Robot Open Autonomous Racing Competition 2020

- Analyzed the system dynamics of a simulated autonomous vehicle and implemented a linear quadratic regulator for control, which markedly improved vehicle stability
- Won Grand Prize in the 2020 ROAR S1 series against six other teams

Environment Monitoring through Microbial Fuel Cells 2018-2020

Prof. Josiah Hester (Northwestern University)

- Analyzed microbial fuel cell data to identify relations between power generation and ambient environmental conditions for persistent battery-less sensing applications

Optical Device Microfabrication 2018-2019

Prof. Hooman Mohseni (Northwestern University)

- Designed an evaluation method for bump bonding; improved yield from 70% to 90%
- Evaluated the effect of reactive ion etching on the BCB polymer used for planarization
- Explored ion-bombardment as a means of large-scale transparent oxide fabrication

Internships

Battery-Sizing for the Reliable Operation of Overloaded DC Microgrids Jun 25-present

Dr. Daniel Gerber (Lawrence Berkeley National Laboratory)

- Calculated minimum battery sizes for the reliable operation of simulated workloads and renewable energies generation on a capacity-constrained residential microgrid
- Designed and implemented a simulation of behind-the-meter DC microgrids in *Python*, comprising battery storage, solar generation, and household loads of EV chargers, HVAC, and large appliances to quantify the relation between battery size, grid capacity, and load profiles and to minimize battery size while reliably supporting all the loads on the microgrid
- Demonstrated a 10kWh battery that reduces required grid capacity from 70 Amps to 15 Amps (80% reduction) by supporting times of peak power in a microgrid containing large appliances
- Evaluated techniques for reducing operating costs and extending battery lifetime
- Simulated a DC microgrid (battery, solar panels, and household appliances) at 1-min resolution over a full year, showed a 10kWh battery could increase solar utilization by 3000kWh annually

Broadband Coverage and Wi-Fi Connectivity for Demand-Adaptive Electrical Loads in Residential Buildings

May 24-Jun 25

Bruce Nordman (Lawrence Berkeley National Laboratory)

- Evaluated Wi-Fi connectivity and broadband availability in the state of California
- Implemented and open-sourcing software to automate finding and forming the network connection between an adaptive load and a local price server over mDNS, which was presented at the 2025 California Demand Flexibility Summit (1 day event), the Home Assistant Community Day & SunSpec Hackathon at Port Labs in Oakland (2 day event).

Deep Neural Networks for Satellites and Atmospheric Science

May-Aug 2023

Johns Hopkins Applied Physics Laboratory

- Evaluated how effective FourCastNet, a recently released deep machine learning model, is at generating predictive models for high-resolution air-quality forecasts
- Evaluated FourCastNet on novel meteorological datasets with different spatial resolutions in three experiments: out-of-the-box, fine-tuned, and fully retrained
- Determined that although the model architecture supports arbitrary spatial resolutions, trained model weights are unable to be applied out-of-the-box on data with different resolutions
- Presented to 50 researchers across different domains at APL as part of the Climate Network research presentations
- Designed and implemented a satellite constellation simulation in Python that used a MATLAB physics engine to determine satellite orbits, positioned targets of interest in locations around the globe, and provided an environment to train a reinforcement agent to select targets within range of each satellite in order to maximize coverage

Computer Vision Summer Internship

May-Aug 2022

Johns Hopkins Applied Physics Laboratory

- Designed and implemented novel neural network models and training methods to incorporate contextual metadata into existing image classification techniques
- Contributed to three computer-vision projects: context-aware image classification, landmine detection on a beach, and ice hazard detection in glacial landscapes
- Annotated images, applied traditional Canny Edge detection techniques, fine-tuned YOLOv8 image-segmentation models using PyTorch Lightning, and implemented the bridge-mode architecture in PyTorch to fine-tuning ResNet18 weights at runtime
- Developed custom ML models, data loaders, data samplers, and loss functions in PyTorch
- Trained with GPUs on remote server clusters
- Improved image classification accuracy by fine-tuning model weights at runtime, in situations with uneven class distribution
- Co-authored a workshop paper in NeurIPS 2022

Bayesian Algorithms for Scanning Probe Microscopy

June-Sept 2019

Dr. Rama K. Vasudevan (Oak Ridge National Laboratory)

- Reprogrammed a Bayesian learning model for parallel deployment on high-performance computing resources for a 150x increase in data analysis throughput

Artificial Intelligence for Enterprise Productivity

June-Sept 2018

Craig Plesco (Ford Motor Company)

- Prototyped, evaluated, and documented four AI projects exploring facial recognition, object recognition, and natural language processing for corporate applications

Leadership

International Graduate Student Ministry

2021-present

President, Financial Agent, and Signatory

- Organizing, publicizing, and reimbursing outreach events to international graduate and professional students to provide support and mentorship in time of transition

Tau Beta Pi Honor Society

2019-2020

Alumni Chair

- Organized and mediated networking and informational events between Tau Beta Pi, the Undergraduate Engineering Office, and other student organizations on campus

Northwestern University Space Technology and Rocketry Club

2016-2017

Ground Station Team Lead

- Managed the design and implementation of communication and visualization programs to receive and display telemetry data transmitted from a rocket's payload

Technical Skills

Programming:	Python, C, C++, MATLAB; Verilog, VHDL; Object-oriented programming
Simulation:	Python, MATLAB
Machine Learning:	PyTorch, TensorFlow; CUDA; Computer Vision (YOLO);
Robotics:	Robot Operating System (ROS2), Linux; kinematics, coordinate transforms, planning, navigation
Controls:	PID, Linear Quadratic Regulator (LQR)
Embedded Systems:	ARM Cortex M4 / M0+; I2C, SPI, GPIO; FreeRTOS; Multi-threading, Finite state machines; sensors (temperature, accelerometer, gyroscope, irradiance, RGB-D camera), memory, drivers
Debugging/Profiling:	Oscilloscope, Logic analyzer; Root cause analysis
Networking:	Bluetooth (BLE), Wi-Fi, Ethernet; HTTP (REST), mDNS, TCP/IP
Knowledge Sharing:	8 publications, 5 presentations, 5 open-source code releases, 2 hardware demos, 1 short film
Tools:	MS Design Suite, Overleaf/LaTeX, GitHub/git
Skills:	Project planning & coordination, Design of experiments, Data analysis, Machine learning, Manuscript/report writing, Leadership and management

Outreach

Bay Area Scientists Inspiring Students (BASIS)

2021-2023

Team Member

- Led virtual electromagnetism and circuit lessons for elementary school students to increase student interest in STEM and higher education

Tau Beta Pi MindSET Program

2020

Program Coordinator

- Coordinated science and technology outreach events between engineering student organizations at Northwestern and local elementary and middle schools

Graduate Coursework

Fall 2022	Teaching Techniques for Electrical Engineering; Slow Cinema
Spring 2022	Applied Behavioral Economics; Rhetoric of Science, Narrative, and Image
Fall 2021	Advanced Topics in Computer Systems; Statistical Learning Theory
Spring 2021	Introduction to Machine Learning; Optimization Models in Engineering
Fall 2020	Linear System Theory; Introduction to Robotics